

# Prerit Sangwan

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2nd-year Computer Science (AI/ML) student with hands-on data science projects in classification and recommendation using pandas and scikit-learn.

## EDUCATION

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**Bennett University**, India — B.Tech, Computer Science and Engineering (AI/ML specialisation) 2nd year, in progress

## PROJECTS

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### Telco Customer Churn Prediction

Jan 2026

*Python, pandas, scikit-learn*

- Prepared the IBM Telco Customer Churn dataset (7,043 customers, cleaned to 7,032): one-hot encoded categorical fields and binned tenure into 12-month groups, giving 50 features.
- Trained a Random Forest classifier (100 trees) to predict churn and saved the model for reuse.

### End-to-End Book Recommendation System

Jun 2026

*Python, pandas, scikit-learn, SciPy, Streamlit*

- Built an item-based collaborative-filtering recommender on the Book-Crossing dataset (1.15M ratings, 271K books).
- Reduced sparsity by keeping users with 200+ ratings and books with 50+ ratings, giving a 742-book by 888-user rating matrix.
- Trained a k-nearest-neighbours model (scikit-learn, sparse CSR input) that returns the 5 most similar books.
- Structured the code as a 4-stage pipeline (ingestion, validation, transformation, training) driven by a YAML config, with custom logging and exception handling; ingestion falls back to local data if the download fails.
- Built a Streamlit app that retrains the model and shows 5 recommendations with cover images; prototyped a cosine-similarity version in Jupyter first.

## SKILLS

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**Language:** Python

**ML and data:** scikit-learn (k-NN, Random Forest), pandas, NumPy, SciPy sparse matrices, collaborative filtering, cosine similarity, one-hot encoding, feature binning

**Tools:** Streamlit, Jupyter, Git and GitHub, YAML-configured modular pipelines, logging and exception handling